Proceedings of the SICE Annual Conference 2017
September 19-22, 2017, Kanazawa University, Kanazawa, Japan

Adaptive Modeling and Compliance Control for RC Servo Motor

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Abstract: This paper concerns compliance motion control for joints of a humanoid robot for the external force. Our strategy is to estimate the amount of the external force by measuring the motor drive current of the RC servo motor since auxiliary current sensors can be attached easily. However, we did not consider about the dynamics of RC servo motor. The aim of this paper is to propose the adaptation scheme to identify the dynamics model of RC servo motor. We construct a time-varying adaptive identifier of the parameters with adaptation law based on adaptive control theory and give successful results in our numerical simulation.

Keywords: Compliance Control, Adaptive Control, Adaptive Identifier

1. INTRODUCTION

Recently, in human society, a robot is working in a daily life of human. The body of such a robot consists of some rigid links and powerful servo motors. The robot joint with an electrical motor is different from the flexible joint of real human. Actually, the joint angles are driven by the powerful motor with high gain feedback law. Therefore, since there is no flexibility for an external force. In order to work then in environment of human, it is necessary to implement the flexibility at the robot joint based on the force control.

In our research, we develop a compliance control system using current sensor for the joint of the humanoid robot (Fig. 2) [1, 2]. The compliance control for the joint of the humanoid robot means the force control against unknown external force. There are many literatures on the compliance control [3-6]. If the compliance control is used in the robot, the robot can prevent a break down of itself by the collision with an object. A performance of the compliance control depends on measurement accuracy of the external force. We suggested the remedy for the problem of measurement accuracy of the external force [2].

In this paper, we propose a method to identify the dynamics model of RC servo motor (Fig. 1). The reason why is we did not consider about the dynamics of RC servo motor in our compliance control system. For that reason, we should be identified the dynamics of RC servo motor. Our new identification system contains the concept of the model reference adaptive control (MRACS). It is well-known that the adaptive control is effective to identify for the dynamics models [7-9]. We also construct a time-varying adaptive identifier to identify the parameters of the model and give successful result for our proposed adaptation scheme. We show usefulness of the adaptive identifier for our research in a numerical simulation.

† Shun Ushida is the presenter of this paper.

Fig. 1 RBS582 product made in JR PROPO

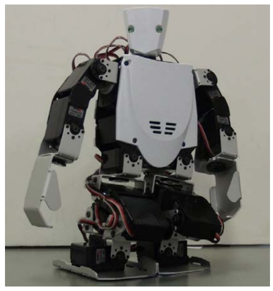

Fig. 2 Robovie-X product made in Vstone

2. PURPOSE OF THE COMPLIANCE CONTROL

2.1. The Object and the Method of Implementation

In our research, we deal with the RC Servo motor (Fig. 2) in humanoid robot. The servo motor has a feedback loop inside and controls position using the feedback scheme. The servo motor receives only the target angle signal, so we can make easily the robotics motion. However, we can not observe the feedback signal of the servo motor, unless we remodel the servo motor.

In order to implement our compliance control to the servo motor, it is necessary to change the motor angle depending on a change of external force. In Fig. 3, we describe a block diagram of our compliance control. we use a current sensor to estimate the external force [10].

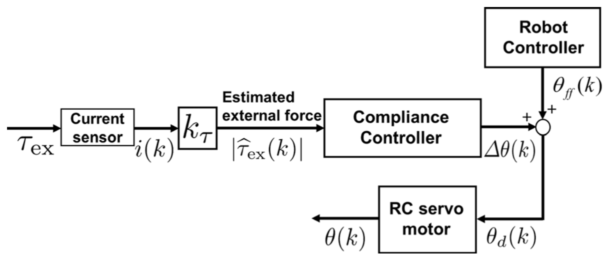

Fig. 3 Block diagram of the compliance control

In order to implement the compliance control for the servo motor, we measure the current i provided by a power supply to the motor, and estimate $\widehat{\tau}_{\rm ex}$ using a toque constant k_τ, where the estimated external force $\hat{\tau}_{\rm ex}$ is given by a following equation

$$\widehat{\tau}_{\rm ex}(k) = k_\tau i(k). \tag{1}$$

The torque constant k_τ can be identified in the experiment in advance. In Fig. 3, we use a microcomputer to implement our compliance control system to the servo motor. Therefore, we deal with the discrete-time system using discretized k in discrete-time.

In next section, we explain about the realization of flexibility of the servo motor using the compliance controller based on Eq. (1).

2.2. The Role of Compliance Controller

We propose the compliance controller that realizes a following equation

$$\widehat{\tau}_{\rm ex}(t) = I_\theta \Delta\ddot{\theta}(t) + D_\theta \Delta\dot{\theta}(t) + K_\theta \Delta\theta(t), \tag{2}$$

where I_θ, D_θ and K_θ are the designable parameters which we can freely choose depending on purpose of robot motion control. The continuous-time different equation given by Eq. (2) is transformed to discrete-time one :

$$\widehat{\tau}_{\rm ex}(k) = I_\theta \Delta\ddot{\theta}(k) + D_\theta \Delta\dot{\theta}(k) + K_\theta \Delta\theta(k), \tag{3}$$

$$\Delta\dot{\theta}(k) = \frac{\Delta\theta(k) - \Delta\theta(k-1)}{T},$$
$$\Delta\ddot{\theta}(k) = \frac{\Delta\dot{\theta}(k) - \Delta\dot{\theta}(k-1)}{T}.$$

Furthermore, we transform Eq. (3) to

$$\begin{aligned} \Delta\theta(k) &= G_1 \widehat{\tau}_{\rm ex} \\ &+ G_2 \Delta\theta(k-1) \\ &+ G_3 \frac{\Delta\theta(k-1) - \Delta\theta(k-2)}{T}, \end{aligned} \tag{4}$$

where T is sampling period of the servo motor, G_1, G_2, G_3 and G_n are coefficients, respectively. $\Delta\theta(k)$ is the modified angle that is determined from designable parameters I_θ, D_θ and K_θ. The flexible joint in humanoid robot can be realized using $\Delta\theta(k)$.

2.3. The Implement of Compliance Control

We implement our compliance control scheme on the control board in Fig. 4. This system consists of one microcomputer and a current sensor. The controlled object of our compliance control shown in Fig. 1 is a servo motor with PWM-type input for positioning system.

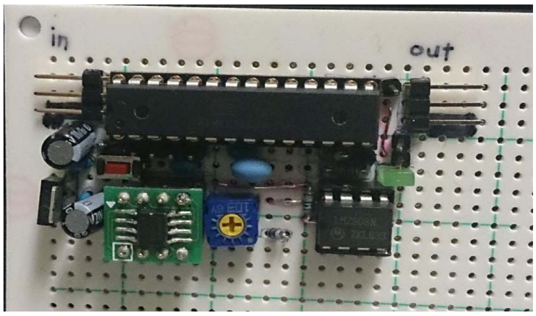

Fig. 4 Compliance Control Board

3. ADAPTIVE SYSTEM FOR COMPLIANCE CONTROL SYSTEM

We have already implemented the compliance control system in RC servo motor. Our system estimates the external force using a current sensor and calculate the output angle to RC servo motor. In the future of our research , we install the compliance control for the real humanoid robot. However, we did not get the model of RC servo motor. Therefore, we suggested the scheme of the parameterized model identification for RC servo motor. If we can get the parameterized model of RC servo motor, we will construct the accurate simulator of compliance control system for humanoid robot and analyze the stability of the system. The identification scheme is the adaptive identifier based on adaptive control system.

3.1. The Parameter Identification of RC Servo Motor Using The Adaptive Identifier

3.1.1. The Parameterized Model on Continuous-Time

We define the liner model of RC servo motor on continuous-time based on [10]. Eq. (5) is the transfer function of RC servo motor on continuous-time.

$$P(s) = \frac{b_3}{s^3 + a_1 s^2 + a_2 s + a_3} \tag{5}$$

thus,

$$\begin{aligned} a_1 &= RJ + K_f L/LJ \\ a_2 &= K_b K_m + K_D K_m/LJ \\ a_3 &= K_p K_m/LJ \\ b_3 &= K_p K_m/LJ. \end{aligned} \tag{6}$$

Where $R[\Omega]$ is armature resistance, L[H] is inductance, K_b[Vs/rad] is counter-electromotive force, J is the mechanical moment of inertia, K_m[Nm/A] is torque coefficient, K_f[Ns/m] is viscous damping coefficient. K_p is proportional gain and K_D is derivative gain in the integrated controller [10].

3.1.2. The Design of Adaptive Identifier on Discrete-Time

Eq. (5) is the transfer function on continuous-time. However, we construct the control system based on micro computer. Therefore, we have to translate into the model of discrete-time from the model of continuous-time. We use the bilinear transform for the transformation of the pulse transfer function. Eq. (7) is the pulse transfer function of RC servo motor.

$$P(z^{-1}) = \frac{\widehat{b}_0 + \widehat{b}_1 z^{-1} + \widehat{b}_2 z^{-2} + \widehat{b}_3 z^{-3}}{1 + \widehat{a}_1 z^{-1} + \widehat{a}_2 z^{-2} + \widehat{a}_3 z^{-3}} \tag{7}$$

the bilinear transform is

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \tag{8}$$

T is sampling time, where $\widehat{a}_1, \widehat{a}_2, \widehat{a}_3, \widehat{b}_0, \widehat{b}_1, \widehat{b}_2$ and $\widehat{b}_3$ are parameters in the model and there are seventh parameters. We have to identify these parameters for the modeling.

3.2. The Adaptive Identifier on Discrete-Time

We use the adaptive identifier for getting the model of RC servo motor. The adaptive identifier is the identification of the parameter based on adaptive control system. There are the effective results in [7-9][11], especially, [11] suggested the design scheme of the adaptive identifier on discrete-time. We construct the adaptive identifier based on [11]. Firstly, we convert into Eq. (9) from Eq. (7).

$$\begin{aligned} A_d(z^{-1})y(k) &= B_d(z^{-1})u(k) \qquad (9)\\ A_d(z^{-1}) &= 1+\sum_{i=1}^{n}\widehat{a}_i z^{-i}\\ B_d(z^{-1}) &= \sum_{i=0}^{n} b_i z^{-i} \end{aligned}$$

$u(k)$ is input signal and $y(k)$ is output signal. The model is 3rd order system and $n = 3$.

3.2.1. The Parameterization for Adaptive Identifier

We convert into Eq. (10) from Eq. (9) for the adaptive identifier. Eq. (10) is linear parameterized model.

$$y(k) = \psi^{\mathrm{T}}\xi(k) \qquad (10)$$

In Eq. (10), we introduce $\eta_i(k)$, $\zeta_i(k)$, σ_i and δ_i. $\eta_i(k)$, $\zeta_i(k)$ are regressors and σ_i, δ_i are the parameters ($i = 0, \cdots, n$).

$$\eta_i(k) = \frac{z^{-i}}{F_d(z^{-1})}u(k) \qquad (11)$$

$$\zeta_i(k) = \frac{z^{-i}}{F_d(z^{-1})}y(k) \qquad (12)$$

$$\sigma = \left[\ \sigma_0, \ \cdots \ , \sigma_i\ \right]^{\mathrm{T}},\ \sigma_i = \widehat{b}_i \qquad (13)$$

$$\delta = \left[\ \delta_0, \ \cdots \ , \delta_i\ \right]^{\mathrm{T}},\ \delta_i = d_i - \widehat{a}_i \qquad (14)$$

$$\eta(k) = \left[\ \eta_0(k), \ \cdots \ , \eta_i(k)\ \right]^{\mathrm{T}} \qquad (15)$$

$$\zeta(k) = \left[\ \zeta_0(k), \ \cdots \ , \zeta_i(k)\ \right]^{\mathrm{T}} \qquad (16)$$

$$\psi = \left[\ \sigma^{\mathrm{T}} \quad \delta^{\mathrm{T}}\ \right]^{\mathrm{T}},\ \xi(k) = \left[\ \eta^{\mathrm{T}}(k) \quad \zeta^{\mathrm{T}}(k)\ \right]^{\mathrm{T}}, \qquad (17)$$

we use $D_d(z^{-1})$ and $F_d(z^{-1})$ for introducing Eq. (10) from Eq. (9). $D_d(z^{-1})$ and $F_d(z^{-1})$ are 3rd order Hurvitz polynomial.

$$D_d(z^{-1}) = 1+\sum_{i=1}^{n} d_i z^{-i} \qquad (18)$$

$$F_d(z^{-1}) = 1+\sum_{i=1}^{n} f_i z^{-i} \qquad (19)$$

we use regressors $\eta_i(k)$, $\zeta_i(k)$ and parameters σ_i, δ_i,

$$\begin{aligned} y(t) &= \frac{F_d(z^{-1})}{D_d(z^{-1})}(\psi^{\mathrm{T}}\xi(k))\\ &= W(z^{-1})(\psi^{\mathrm{T}}\xi(k)). \qquad (20) \end{aligned}$$

If it be so $W(z^{-1})$ as strictly positive real transfer function, we can choice as $F_d(z^{-1}) = D_d(z^{-1})$. Therefore, we can get Eq. (10) from $W(z^{-1}) = 1$.

3.2.2. The Consist of Adaptive Identifier

We propose the block diagram of adaptive identifier for RC servo motor in Fig. 5. In this system, we can use input signal $u(k)$ and output signal $y(k)$. These signals make the regressors $\eta(k)$ and $\zeta(k)$ for the construction of identified model and the identification of the parameters.

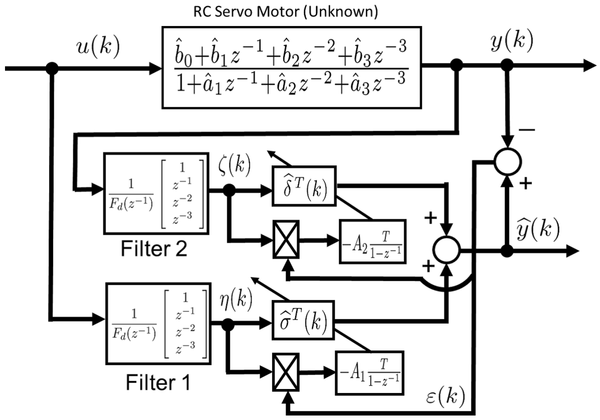

Fig. 5 Adaptive identifier for RC servo motor

$u(k)$ is the excitation signal which has the plural frequency component based on persistently exciting (PE) [8].

3.2.3. The Adaptation Law

We define the adaptation law that is given by following equation. Eq. (21) is a same pattern as Eq. (10), we just replace ψ with $\widehat{\psi}(k)$.

$$\widehat{y}(k) = \widehat{\psi}^{\mathrm{T}}(k)\xi(k) \qquad (21)$$

$\varepsilon(k)$ is the identification error from Eq. (21) - Eq. (10)

$$\varepsilon(k) = \widehat{y}(k) - y(k), \qquad (22)$$

hence, the error equation is

$$\varepsilon(k) = \widetilde{\psi}^{\mathrm{T}}(k)\xi(k), \qquad \widetilde{\psi}(k) = \widehat{\psi}(k) - \psi. \qquad (23)$$

$\widehat{\psi}(k)$ is the variable parameters. We use the adaptation law Eq. (24) to identify these parameters based on Eq. (23).

$$\widehat{\psi}(k) = \widehat{\psi}(k-1) - TA\xi(k)\varepsilon(k) \qquad (24)$$

$$A = A^{\mathrm{T}} > 0, \quad A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \qquad (25)$$

If $\varepsilon(k) \to 0$, the update of these parameters are finished and converge a constant value. A is the adaptation gain (positive definite matrix).

3.3. Numerical Simulation and Results

We tried on the simulation to conform the effectiveness of the adaptive identifier. In additional, we explain about setting and results.

3.3.1. Setting

Set values are

$$\begin{cases} \widehat{a}_0 = 1, & \widehat{b}_0 = 0.003618 \\ \widehat{a}_1 = -1.519, & \widehat{b}_1 = 0.01085 \\ \widehat{a}_2 = 0.6652, & \widehat{b}_2 = 0.01085 \\ \widehat{a}_3 = -0.1174, & \widehat{b}_3 = 0.003618, \end{cases} \tag{26}$$

the coefficient of filter $F_d(z^{-1})$ are

$$f_1 = 0.01,\ f_2 = 0.05,\ f_3 = 0.01, \tag{27}$$

the adaptation gain is

$$A_1 = A_2 = \begin{pmatrix} 1.0 \times 10^4 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & 1.0 \times 10^4 \end{pmatrix}. \tag{28}$$

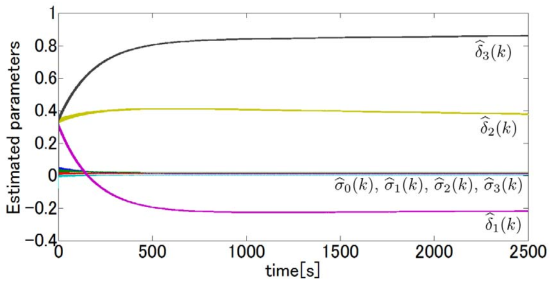

Fig. 6 The identification of the parameters

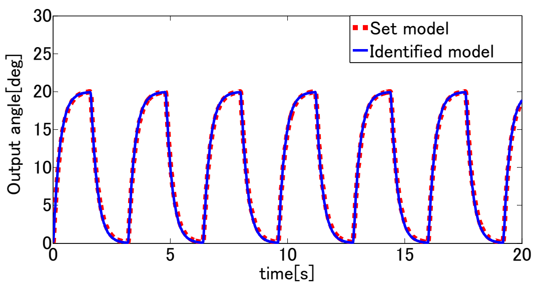

Fig. 7 The output of the identified model

In Fig. 6, seventh parameters $\widehat{\sigma}_0(k)$-$\widehat{\sigma}_3(k)$, $\widehat{\delta}_1(k)$-$\widehat{\delta}_3(k)$ are converge to constant value, and then we conform that identified model accords with set model from Fig. 7 using these constant values. Therefore, we were giving the successful result on the simulation for the identification of RC servo motor.

4. CONCLUSION

In this paper, we pointed out that the identification of RC servo motor for our compliance control. In order to identify the identified model, we constructed the adaptive identifier using a scheme of adaptive control. In the simulation, the adaptive identifier has been successfully applied to the identification of RC servo motor.

The next step of our development will be implement the adaptive identifier for RC servo motor. On the other hands, if we can get the model of RC servo motor, we construct the accurate simulator of our system.

REFERENCES

[1] K. Shibata, M. Hayashi, Y. Ikeda, S. Ushida, H. Oku, "Compliance Control for RC Servo Motor Based on a Simple External Force Estimate Using Current Sensor," *The 59th Annual Conference of the Institute of Systems, Control and Information Engineers, 2015.*

[2] M. Hayashi, K. Shibata, S. Ushida, H. Oku and W. Kongprawechnon, "Compliance Control of the RC Servo Motor with Adarptive Identifier", *Proc. of the SICE Annual Conference 2016, Tsukuba, Japan, September 20-23*, 2016.

[3] K. Shimura, Y. Hori, "Robustified Force and Collision Control for Robot Manipulator," *Journal of the Robotics Society of Japan*, Vol. 11, No. 2, pp. 235-245, 1993.

[4] S. Ando, R. Nagai, Y. Inoue, "Automatic Damping Tuning of Impedance Control for Assembly Task," *Journal of the Robotics Society of Japan*, Vol. 29, No. 7, pp. 564-572, 2011.

[5] T. Umeno, Y. Hori, "Design of Robust servo systems Based on the Parametrization of Two Degrees of Freedom Control Systems," *IEEJ Transactions on Industry Applications*, Vol. 109, No. 11, pp. 825-832, 1989.

[6] T. Umeno, Y. Hori, "Design of the Two Degrees of Freedom Robust Servo system and its Application to Motion control of Robot Manipulators," *IEEJ Transactions on Industry Applications*, Vol. 110, No. 11, pp. 1163-1170, 1990.

[7] K. Tanaka and T. Suzuki, "Consideration on Robustness of Continuous-Time Adaptive Identifiers", *Transactions of the Society of Instrument and Control Engineers*, Vol. 21, No. 11, pp. 1156-1162, 1985.

[8] T. Suzuki, "*Adaptive Control*," CORONA PUBLISHING CO.LTD, 2001.

[9] K. J. Åstrom, B. Wittenmark, "*Adaptive Control, Second Edition*," Addison Wesley, 1995.

[10] M. Ishikawa, R. Kitayoshi, T. Wada, I. Maruta, T. Sugie, "Modeling of R/C Servo Motor and Application to Underactuated Mechanical Systems," *Transactions of the Society of Instrument and Control Engineers*, Vol. 46, No. 4, pp. 237-244, 2010.

[11] K. Ichikawa, "*Adaptive Control*," SHOKODO CO.LTD, 1984.